Lab11

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##Class 11 Lab

```
# Change this for YOUR results dir name
results_dir <- "HIVPRDimer_23119_0"

pdb_files <- list.files(path=results_dir,
                        pattern="*.pdb",
                        full.names = TRUE)

# Print our PDB file names
basename(pdb_files)
```

```
[1] "HIVPRDimer_23119_0_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb"
[2] "HIVPRDimer_23119_0_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb"
[3] "HIVPRDimer_23119_0_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb"
[4] "HIVPRDimer_23119_0_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb"
[5] "HIVPRDimer_23119_0_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb"
```

```
library(bio3d)

# Read all data from Models and superpose/fit coords
pdbs <- pdbaln(pdb_files, fit=TRUE, exefile="msa")
```

```
Reading PDB files:
HIVPRDimer_23119_0/HIVPRDimer_23119_0_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed
HIVPRDimer_23119_0/HIVPRDimer_23119_0_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed
HIVPRDimer_23119_0/HIVPRDimer_23119_0_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed
HIVPRDimer_23119_0/HIVPRDimer_23119_0_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed
HIVPRDimer_23119_0/HIVPRDimer_23119_0_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed
.....
```

```
Extracting sequences

pdb/seq: 1   name: HIVPRDimer_23119_0/HIVPRDimer_23119_0_unrelaxed_rank_001_alphafold2_multim
pdb/seq: 2   name: HIVPRDimer_23119_0/HIVPRDimer_23119_0_unrelaxed_rank_002_alphafold2_multim
pdb/seq: 3   name: HIVPRDimer_23119_0/HIVPRDimer_23119_0_unrelaxed_rank_003_alphafold2_multim
pdb/seq: 4   name: HIVPRDimer_23119_0/HIVPRDimer_23119_0_unrelaxed_rank_004_alphafold2_multim
pdb/seq: 5   name: HIVPRDimer_23119_0/HIVPRDimer_23119_0_unrelaxed_rank_005_alphafold2_multim
```

```
pdbs
```

```
                              1        .         .         .         .         50
[Truncated_Name:1]HIVPRDimer  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:2]HIVPRDimer  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:3]HIVPRDimer  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:4]HIVPRDimer  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:5]HIVPRDimer  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
                              **************************************************
                              1        .         .         .         .         50

                             51        .         .         .         .         100
[Truncated_Name:1]HIVPRDimer  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:2]HIVPRDimer  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:3]HIVPRDimer  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:4]HIVPRDimer  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:5]HIVPRDimer  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
                              **************************************************
                             51        .         .         .         .         100

                            101        .         .         .         .         150
[Truncated_Name:1]HIVPRDimer  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:2]HIVPRDimer  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:3]HIVPRDimer  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:4]HIVPRDimer  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:5]HIVPRDimer  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
                              **************************************************
                            101        .         .         .         .         150

                            151        .         .         .         .       198
[Truncated_Name:1]HIVPRDimer  GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:2]HIVPRDimer  GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:3]HIVPRDimer  GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
```

```
[Truncated_Name:4]HIVPRDimer   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:5]HIVPRDimer   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
                               **********************************************
                             151        .         .         .         .       198

Call:
  pdbaln(files = pdb_files, fit = TRUE, exefile = "msa")

Class:
  pdbs, fasta

Alignment dimensions:
  5 sequence rows; 198 position columns (198 non-gap, 0 gap)

+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

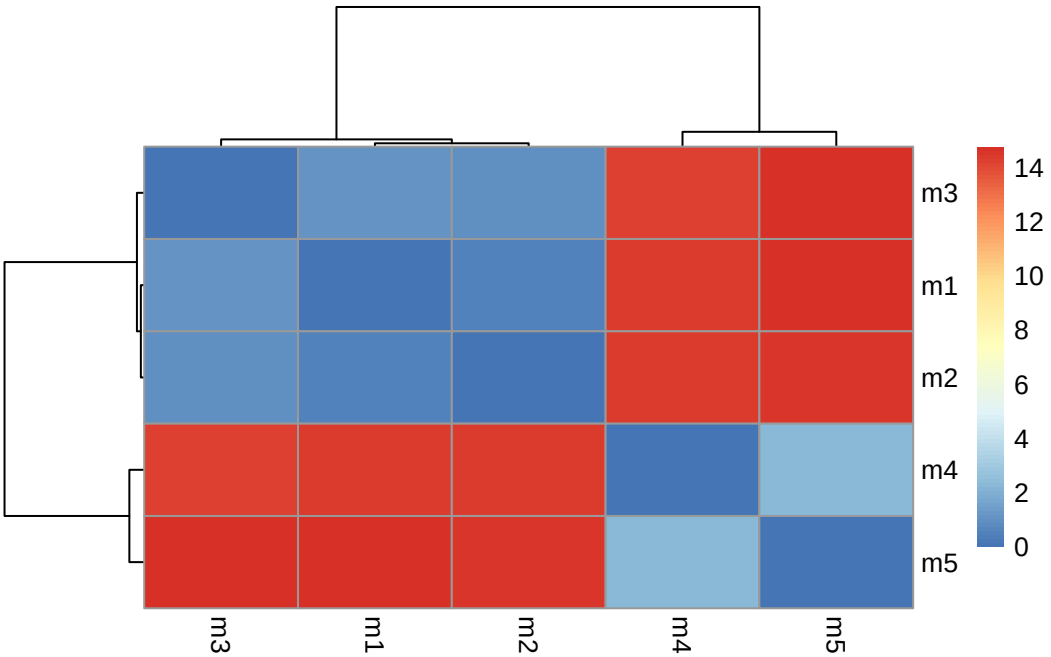
```r
rd <- rmsd(pdbs, fit=T)
```

```
Warning in rmsd(pdbs, fit = T): No indices provided, using the 198 non NA positions
```

```r
range(rd)
```

```
[1]  0.000 14.754
```

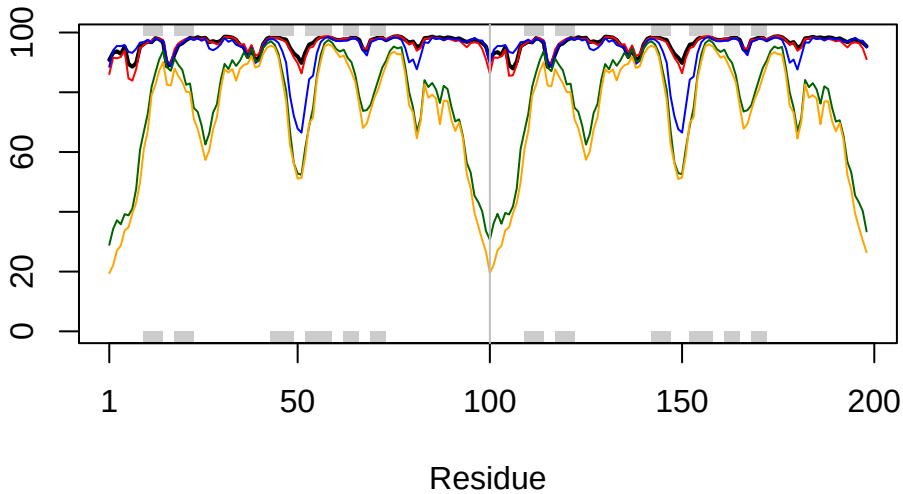
```r
library(pheatmap)

colnames(rd) <- paste0("m",1:5)
rownames(rd) <- paste0("m",1:5)
pheatmap(rd)
```




```
# Read a reference PDB structure
pdb <- read.pdb("1hsg")
```

```
  Note: Accessing on-line PDB file
```

```
plotb3(pdbs$b[1,], typ="l", lwd=2, sse=pdb)
points(pdbs$b[2,], typ="l", col="red")
points(pdbs$b[3,], typ="l", col="blue")
points(pdbs$b[4,], typ="l", col="darkgreen")
points(pdbs$b[5,], typ="l", col="orange")
abline(v=100, col="gray")
```




```r
core <- core.find(pdbs)
```

```
core size 197 of 198  vol = 9885.822
core size 196 of 198  vol = 6896.71
core size 195 of 198  vol = 1337.847
core size 194 of 198  vol = 1040.67
core size 193 of 198  vol = 951.857
core size 192 of 198  vol = 899.083
core size 191 of 198  vol = 834.732
core size 190 of 198  vol = 771.338
core size 189 of 198  vol = 733.065
core size 188 of 198  vol = 697.28
core size 187 of 198  vol = 659.742
core size 186 of 198  vol = 625.273
core size 185 of 198  vol = 589.541
core size 184 of 198  vol = 568.253
core size 183 of 198  vol = 545.015
core size 182 of 198  vol = 512.889
core size 181 of 198  vol = 490.723
core size 180 of 198  vol = 470.266
core size 179 of 198  vol = 450.731
core size 178 of 198  vol = 434.735
```

```
core size 177 of 198  vol = 420.337
core size 176 of 198  vol = 406.658
core size 175 of 198  vol = 393.334
core size 174 of 198  vol = 382.395
core size 173 of 198  vol = 372.858
core size 172 of 198  vol = 356.994
core size 171 of 198  vol = 346.567
core size 170 of 198  vol = 337.446
core size 169 of 198  vol = 326.659
core size 168 of 198  vol = 314.95
core size 167 of 198  vol = 304.127
core size 166 of 198  vol = 294.552
core size 165 of 198  vol = 285.648
core size 164 of 198  vol = 278.884
core size 163 of 198  vol = 266.765
core size 162 of 198  vol = 258.994
core size 161 of 198  vol = 247.723
core size 160 of 198  vol = 239.84
core size 159 of 198  vol = 234.963
core size 158 of 198  vol = 230.062
core size 157 of 198  vol = 221.985
core size 156 of 198  vol = 215.62
core size 155 of 198  vol = 206.793
core size 154 of 198  vol = 196.984
core size 153 of 198  vol = 188.539
core size 152 of 198  vol = 182.262
core size 151 of 198  vol = 176.954
core size 150 of 198  vol = 170.712
core size 149 of 198  vol = 166.119
core size 148 of 198  vol = 159.796
core size 147 of 198  vol = 153.767
core size 146 of 198  vol = 149.092
core size 145 of 198  vol = 143.657
core size 144 of 198  vol = 137.138
core size 143 of 198  vol = 132.517
core size 142 of 198  vol = 127.231
core size 141 of 198  vol = 121.574
core size 140 of 198  vol = 116.775
core size 139 of 198  vol = 112.57
core size 138 of 198  vol = 108.17
core size 137 of 198  vol = 105.133
core size 136 of 198  vol = 101.249
core size 135 of 198  vol = 97.374
```

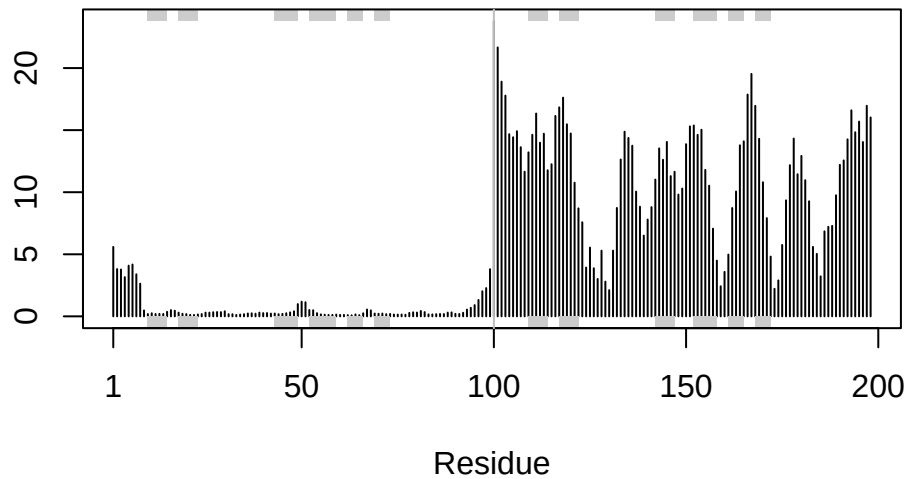
```
core size 134 of 198  vol = 92.974
core size 133 of 198  vol = 88.184
core size 132 of 198  vol = 84.029
core size 131 of 198  vol = 81.898
core size 130 of 198  vol = 78.019
core size 129 of 198  vol = 75.272
core size 128 of 198  vol = 73.052
core size 127 of 198  vol = 70.695
core size 126 of 198  vol = 68.975
core size 125 of 198  vol = 66.694
core size 124 of 198  vol = 64.394
core size 123 of 198  vol = 62.092
core size 122 of 198  vol = 59.045
core size 121 of 198  vol = 56.629
core size 120 of 198  vol = 54.016
core size 119 of 198  vol = 51.806
core size 118 of 198  vol = 49.652
core size 117 of 198  vol = 48.193
core size 116 of 198  vol = 46.648
core size 115 of 198  vol = 44.752
core size 114 of 198  vol = 43.292
core size 113 of 198  vol = 41.093
core size 112 of 198  vol = 39.147
core size 111 of 198  vol = 36.472
core size 110 of 198  vol = 34.117
core size 109 of 198  vol = 31.47
core size 108 of 198  vol = 29.448
core size 107 of 198  vol = 27.325
core size 106 of 198  vol = 25.822
core size 105 of 198  vol = 24.15
core size 104 of 198  vol = 22.648
core size 103 of 198  vol = 21.069
core size 102 of 198  vol = 19.953
core size 101 of 198  vol = 18.3
core size 100 of 198  vol = 15.723
core size 99 of 198  vol = 14.841
core size 98 of 198  vol = 11.646
core size 97 of 198  vol = 9.434
core size 96 of 198  vol = 7.354
core size 95 of 198  vol = 6.179
core size 94 of 198  vol = 5.666
core size 93 of 198  vol = 4.705
core size 92 of 198  vol = 3.665
```

```
 core size 91 of 198  vol = 2.77
 core size 90 of 198  vol = 2.151
 core size 89 of 198  vol = 1.715
 core size 88 of 198  vol = 1.15
 core size 87 of 198  vol = 0.874
 core size 86 of 198  vol = 0.685
 core size 85 of 198  vol = 0.528
 core size 84 of 198  vol = 0.37
 FINISHED: Min vol ( 0.5 ) reached
```

```r
core.inds <- print(core, vol=0.5)
```

```
# 85 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1     9  49     41
2    52  95     44
```

```r
xyz <- pdbfit(pdbs, core.inds, outpath="corefit_structures")



rf <- rmsf(xyz)

plotb3(rf, sse=pdb)
abline(v=100, col="gray", ylab="RMSF")
```



##Predicted Alignment for Error:

```
library(jsonlite)

# Listing of all PAE JSON files
pae_files <- list.files(path=results_dir,
                        pattern=".*model.*\\.json",
                        full.names = TRUE)

pae1 <- read_json(pae_files[1],simplifyVector = TRUE)
pae5 <- read_json(pae_files[5],simplifyVector = TRUE)

attributes(pae1)
```

```
$names
[1] "plddt"   "max_pae" "pae"     "ptm"     "iptm"
```

```
# Per-residue pLDDT scores
#  same as B-factor of PDB..
head(pae1$plddt)
```

```
[1] 90.81 93.25 93.69 92.88 95.25 89.44
```

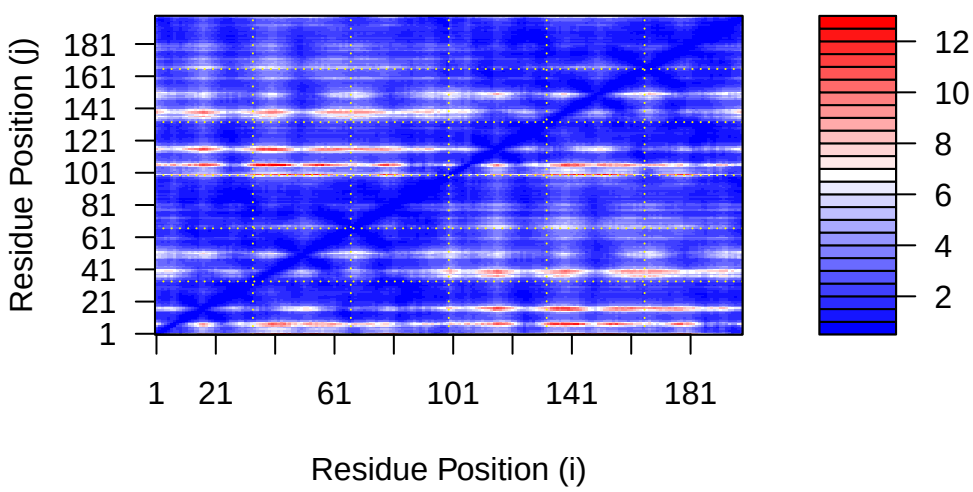
```r
pae1$max_pae
```

```
[1] 12.84375
```

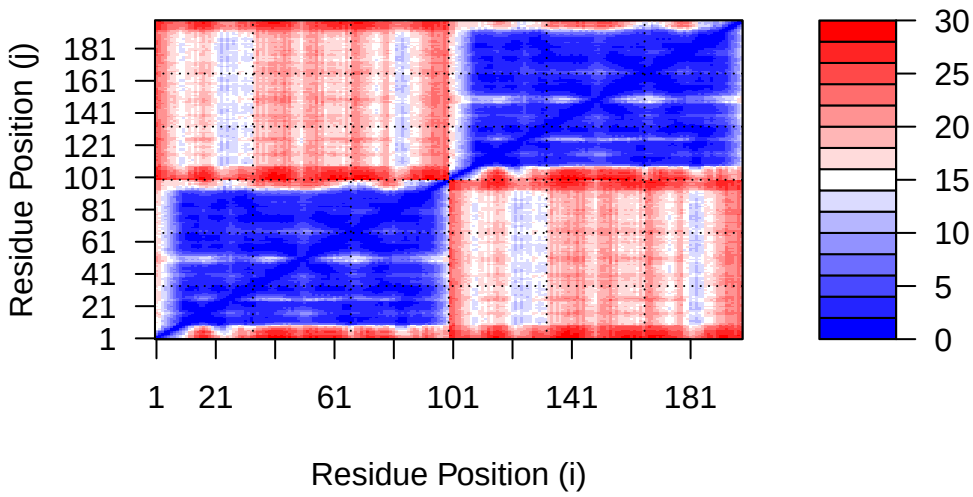
```r
pae5$max_pae
```

```
[1] 29.59375
```

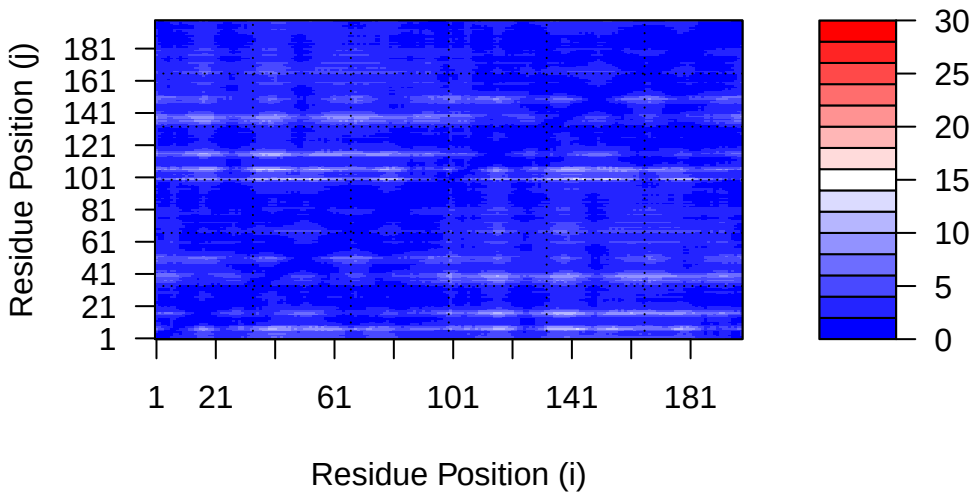
```r
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)")
```




```r
plot.dmat(pae5$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```




```
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```




```
aln_file <- list.files(path=results_dir,
                       pattern=".a3m$",
                        full.names = TRUE)
aln_file
```

```
[1] "HIVPRDimer_23119_0/HIVPRDimer_23119_0.a3m"
```

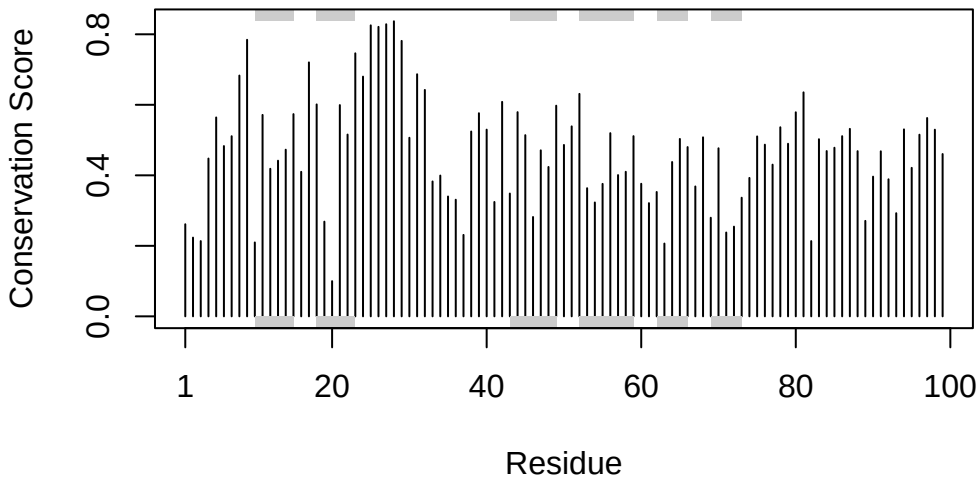
```
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

```
[1] " ** Duplicated sequence id's: 101 **"
[2] " ** Duplicated sequence id's: 101 **"
```

```
dim(aln$ali)
```

```
[1] 5397  132
```

```
sim <- conserv(aln)

plotb3(sim[1:99], sse=trim.pdb(pdb, chain="A"),
       ylab="Conservation Score")
```




```
con <- consensus(aln, cutoff = 0.9)
con$seq
```

```
  [1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-" "-"
 [37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```

```
m1.pdb <- read.pdb(pdb_files[1])
occ <- vec2resno(c(sim[1:99], sim[1:99]), m1.pdb$atom$resno)
write.pdb(m1.pdb, o=occ, file="m1_conserv.pdb")
```